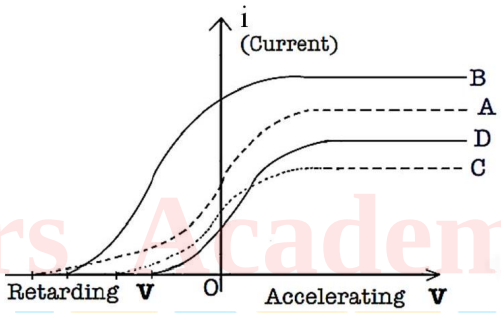
SR.IIT *(MODEL-A)&ALL IIT **Date: 18-01-26**
Time: 3 Hrs **SPL GTM TEST-4** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

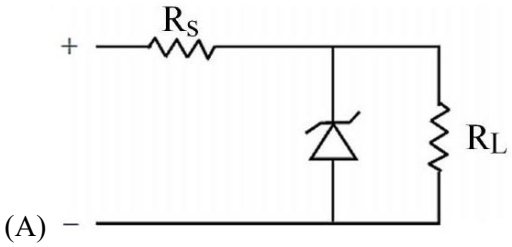
This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

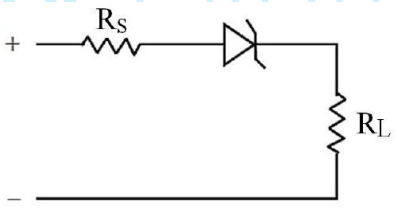
1. Figure shows the results of an experiment involving photoelectric effect. The graphs A, B, C, D corresponds to the light beam having different wavelengths. Choose the incorrect statement.



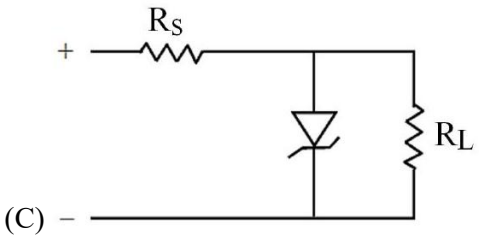
(A) Beam A has lowest frequency
(B) Beam D has longest wavelength
(C) Beam B has the highest rate of photoelectric emission
(D) Photoelectrons ejected by beam A have the highest momentum

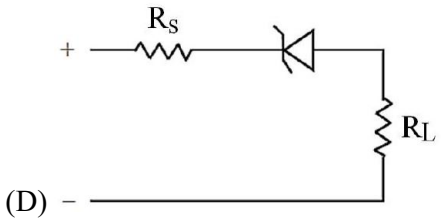
2. A zener diode is to be used as a voltage regulator. Identify the correct set up.





For More Join: @IITJEE_Advanced





3. At distance of 10 cm and 15 cm radially outwards from the surface of a uniformly charged solid sphere, the potentials are 100 V and 75 V respectively. Then choose correct statement.

(A) Potential at its surface is 200 V

(B) The charge on the sphere is $(5/3)\times10^{-10}$ C

(C) The electric field on the surface is $2000\ V/m$

(D) The electric potential at its centre is 450 V

4. In a resonance tube experiment, a closed organ pipe of length 150 cm resonates when tuned with a tuning fork of frequency 340 Hz . If water is poured in the pipe, then (given $v_{air} = 340\ m/sec.$) (Ignore end correction)

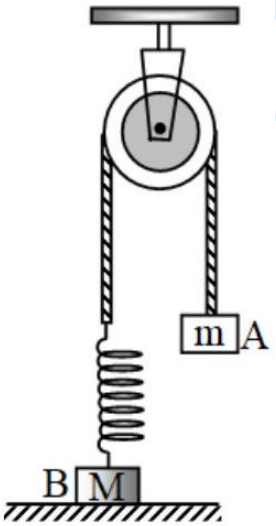
(A) minimum length of water column to have the resonance is 55 cm

(B) the distance between two successive nodes is 50 cm

(C) the maximum length of water column to create the resonance is 100 cm

(D) none of these

5. In the figure the block A is released from rest when the spring is at its natural length. For the block B of mass $M = 20$ kg to leave contact with the ground at some stage, the minimum mass of A must be [Assume the spring, string & pulley to be ideal]



For More Join: @IITJEE_Advanced

(A) 40 kg

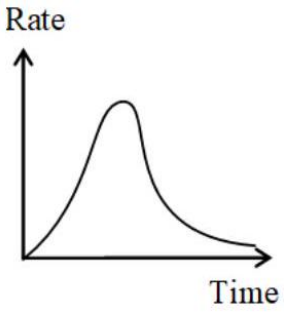
(B) 20 kg

(C) 10 kg

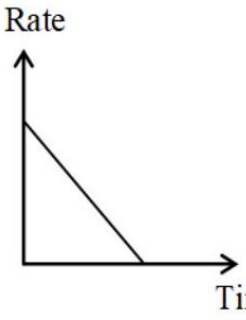
(D) a function of M and the force constant of the spring

6. In a LR circuit connected to a battery the rate at which energy is stored in the inductor is plotted against time during the growth of current in the circuit. Which of the following best represents the resulting curve ?

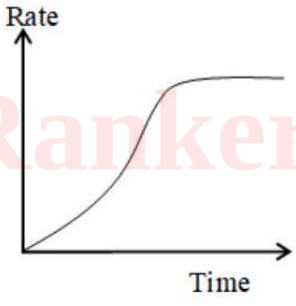
(A)



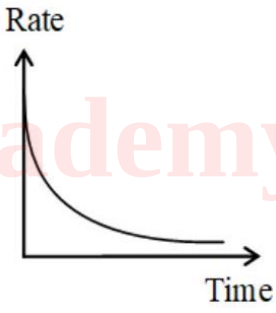
(B)



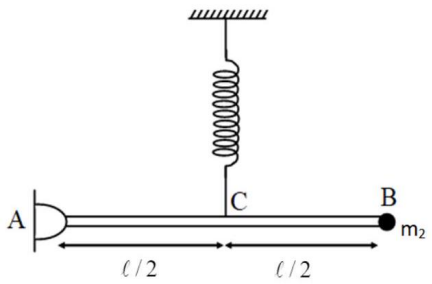
(C)



(D)



7. A uniform thin rod having length ℓ has a mass m_1 kg and carries a mass m_2 kg at B . The rod is hinged at A and is maintained in the horizontal position by a spring having a spring constant k at C as shown in figure. Find the time period of oscillation of given system for oscillation in vertical plane.



For More Join: @IITJEE_Advanced

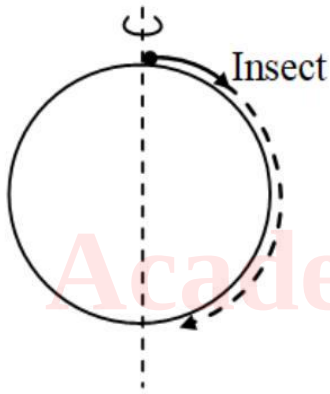
(A) $2\pi\sqrt{\frac{m_1+3m_2}{3k}}$ (B) $2\pi\sqrt{\frac{3m_1+m_2}{3k}}$

(C) $4\pi\sqrt{\frac{m_1+3m_2}{3k}}$ (D) $4\pi\sqrt{\frac{3m_1+m_2}{3k}}$

8. A potential difference of 200 V is applied to a coil at a temperature of $15^\circ C$ and the current is 10 A . What will be the temperature of the coil when the current has fallen to 5 A , the applied voltage being the same as before ? (Given $\alpha = \frac{1}{234}C^{-1}$ at $0^\circ C$)

(A) $364^\circ C$ (B) $164^\circ C$ (C) $464^\circ C$ (D) $264^\circ C$

9. A solid sphere is rotating about an axis as shown in figure. An insect follows the dotted path on the surface of sphere relative to the sphere as shown. Match the following :



	Column-1		Column-2
(A)	Moment of inertia	(P)	will remain constant
(B)	Angular velocity	(Q)	will first increase then decrease
(C)	Angular momentum	(R)	will first decrease then increase
(D)	Rotational kinetic energy	(S)	will continuously decrease

(A) $A \to S; B \to P; C \to P; D \to R$ (B) $A \to Q; B \to R; C \to P; D \to R$

(C) $A \to P; B \to R; C \to S; D \to R$ (D) $A \to Q; B \to S; C \to P; D \to P$ 10.

10. Which statement is correct?

(A) N-type germanium is negatively charged and P-type germanium is positively charged

(B) Both N-type and P-type germanium are neutral

(C) N-type germanium is positively charged and P-type germanium is negatively charged

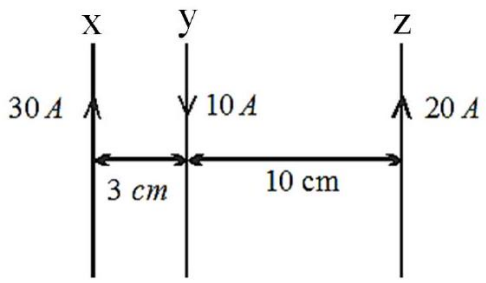
(D) Both N-type and P-type germanium are negatively charged

For More Join: @IITJEE_Advanced

11. A small bar magnet placed with its axis at 30° with an external field of 0.06 T experiences a torque of 0.018 Nm . The minimum work required to rotate it from its stable to unstable equilibrium position is -

(A) 9.2×10^{-3} J (B) 6.4×10^{-2} J (C) 11.7×10^{-3} J (D) 7.2×10^{-2} J

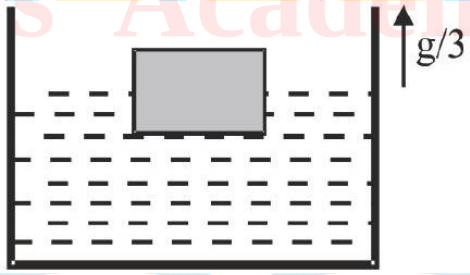
12.. In the figure, x, y and z are long straight wires. The magnetic force on 25 cm length of y is



(A) 2×10^{-4} N towards right (B) 4×10^{-4} N towards right

(C) 2×10^{-4} N towards left (D) 4×10^{-4} N towards left

13. A cubical block is floating in a liquid with half of its volume immersed in the liquid. When the whole system accelerates upwards with acceleration of $g/3$, the fraction of volume immersed in the liquid will be :-



(A) $\frac{1}{2}$ (B) $\frac{3}{8}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$

14. In the laboratory, diffraction of light by a single slit is being observed. If slit is made slightly narrow, then diffraction pattern will :

(A) be more spreaded than before (B) be less spreaded than before

(C) be spreaded as before (D) be disappeared

15. **Statement-I** : In Young's double slit experiment, we observe an interference pattern on the screen if both the slits are illuminated by two bulbs of same power.

Statement-II : The interference pattern is observed when source is monochromatic and coherent.

(A) Both Statement-I and Statement-II are true, and Statement-II is the correct explanation of Statement-I.

Rankers Academy JEE
THE NARAYANA GROUP
For More Join: @IITJEE_Advanced

(B) Both Statement-I and Statement-II are true but Statement-II is not the correct explanation of Statement-I.

(C) Statement-I is true but Statement-II is false.

(D) Statement-I is false but Statement-II is true

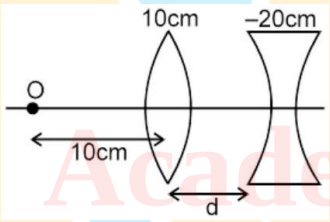
16. The angle of incidence at which reflected light is totally polarized for reflection from air to glass (refractive index n)

(A) $\sin^{-1}(n)$ (B) $\sin^{-1}\left(\frac{1}{n}\right)$ (C) $\tan^{-1}\left(\frac{1}{n}\right)$ (D) $\tan^{-1}(n)$

17. A double-slit arrangement produces interference fringes that are 0.25° apart using wavelength 600 nm . If angular separation is increased to 0.3° by changing wavelength of the light used, then the new wavelength must be :

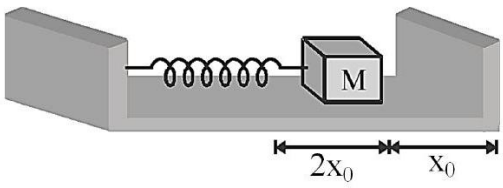
(A) 480 nm (B) 540 nm (C) 720 nm (D) 840 nm

18. What should be the value of distance d so that final image is formed on the object itself ? (focal lengths of the lenses are as given in the figure).



(A) 10 cm (B) 20 cm (C) 5 cm (D) none of these

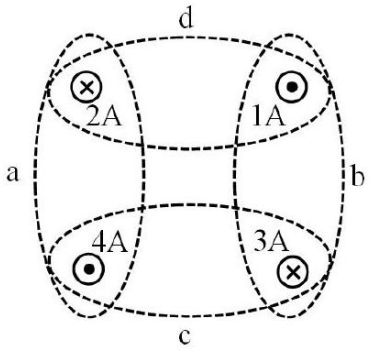
19. One end of a spring of force constant k is fixed to a vertical wall and the other to a block of mass m resting on a smooth horizontal surface. There is another wall at a distance x_0 from the block. The spring is then compressed by $2x_0$ and released. The time taken to strike the wall is:



(A) $\frac{1}{6}\pi\sqrt{\frac{m}{k}}$ (B) $\sqrt{\frac{m}{k}}$ (C) $\frac{2\pi}{3}\sqrt{\frac{m}{k}}$ (D) $\frac{\pi}{4}\sqrt{\frac{m}{k}}$

For More Join: @IITJEE_Advanced

20. Value of $\oint B.d\ell$ is :-



(A) a = b > c > d (B) $(a=b)<(c=d)$

(C) $(a=b)>(c=d)$ (D) None of these

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

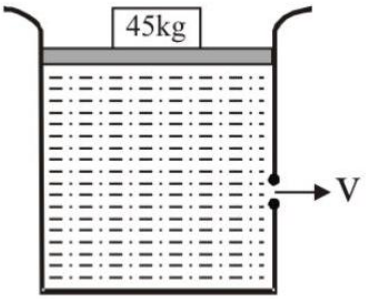
21. A uniform magnetic field exists in region given by $\vec{B}=\left(3\hat{i}+4\hat{j}+5\hat{k}\right)T$. A rod of length 5 m is placed along y -axis is moved along x -axis with constant speed 1 m / sec. Find out induced emf (in V) in the rod.

22. An electron is accelerated with potential v_0 and it collides with a stationary helium ion (He^+) in its ground state and excites to a higher level. After the collision, He^+ ions emits two photons in succession with wavelength 1085Å and 304Å. If the energy of the electron after the collision is 47.7 eV , then find the value of $\frac{V_0}{n}$ where n is the excitation state of electron in He^+ due to collision.

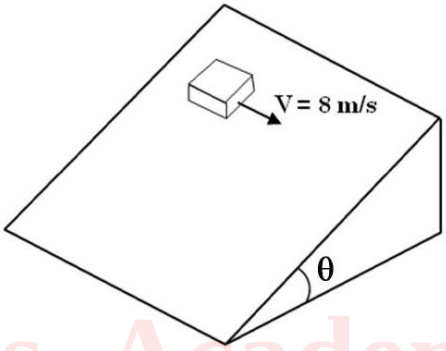
Given $hc=12431eV\times A^0$.

23. A large cylindrical tank of cross-sectional area 1 m^2 is filled with water. It has a small hole at a height of 1 m from the bottom. A movable piston of mass 5 kg is fitted on the top of the tank such that it can slide in the tank freely. A load of 45 kg is applied on the top of water by piston, as shown in figure. Find the value of $v(\text{in m/s})$ when piston is 7 m above the bottom. $\left(g=10\ m/s^2\right)$

For More Join: @IITJEE_Advanced



24. A block is placed on a plane inclined at angle θ. The coefficient of friction between the block and plane is $\mu = \tan\theta$. The block is given a kick so that it initially moves with speed $8\ \text{m/s}$ horizontally (i.e., in the direction perpendicular to the direction pointing straight down the plane; see the figure). What is the speed of the block (in m/s) after a very long time?



25. A planet revolves about the sun in elliptical orbit of semi-major axis 2×10^{12} m. The areal velocity of the planet when it is nearest to the sun is $4.4\times10^{16}\ \text{m/s}$. The least distance between planet and the sun is 1.8×10^{12} m. The minimum speed of the planet in km/s is k. Determine the value of k.

CHEMISTRY MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. Which statement among the following is incorrect?
(A) Helium is present in radioactive mineral cleveite
(B) The first ionization enthalpy of molecular oxygen is almost the same as that of xenon
(C) BiF_5 is the only well-characterized compound of bismuth in the +5 oxidation state
(D) In acidic conditions, nitrogen compounds (only in the +3 oxidation state) exhibits disproportionation but other oxidation state are stable in acidic medium.

27. Which of the following statement correctly describes the behaviour of the permanganate ion $\left(MnO_4^-\right)$ under different conditions?
(A) The standard reduction potential for MnO_4^- to MnO_4^{2-} is higher than that for its reduction to

For More Join: @IITJEE_Advanced

Mn^{2+}, making MnO_4^{2-} the preferred product in acidic conditions.

(B) In neutral or slightly basic solutions, permanganate oxidizes iodide ions to periodate.

(C) Permanganate converts Mn^{2+} (from manganous salts) to MnO_2, a reaction that is catalyzed by zinc sulphate or zinc oxide in slightly basic medium.

(D) At $\left[H^+\right] = 1M$, permanganate rapidly oxidizes water without the need for Mn^{2+} ions or low temperatures, owing to its high redox potential.

28. **Assertion (A)** : In a period of the periodic table, chemical reactivity is highest at the two extremes and lowest at the centre (exclude noble gases).
Reason (R) : The high reactivity at the left extreme of a period is due to the ease of electron loss resulting from low ionization enthalpy and in case of right extreme high electron affinity.
(A) Both Assertion and Reason are true, and the Reason is the correct explanation of the Assertion
(B) Both Assertion and Reason are true, but the Reason is not the correct explanation of the Assertion
(C) Assertion is true, but Reason is false.
(D) Assertion is false, but Reason is true.

29. Which of the following correctly represents the order of the specified property for the given molecules or ions?

(A) Bond length: N=O $\langle C = C \rangle C = N$

(B) Dipole moment: $H_2O > NH_3 > NF_3$

(C) Bond angle: $NO_2 > NO_2^- > NO_2^+$

(D) Maximum number of atoms in a plane: $H_2O > CO_2 < BF_3$

30. Match List-I (Ions) with List-II (Confirmation Tests)

	List-I (Ion)		List-II (Confirmation Test Description)
(P)	Nitrite (NO_2^-)	1.	Sample $+K_2Cr_2O_7 +$ conc. $H_2SO_4 \rightarrow$ Coloured Gas Coloured Gas $+NaOH \rightarrow$ yellow solution yellow solution $+CH_3COOH + Pb(CH_3COO)_2 \rightarrow$ yellow ppt.
(Q)	Chloride (Cl)	2.	Sample $+CH_3COOH +$ Sulphanilicacid $+ NaNO_2 + HCl$ (cold) $\rightarrow$ Diazonium salt Diazonium salt + 1-Naphthylamine $\rightarrow$ Intense red azo dye
(R)	Acetate (CH_3COO^-)	3.	Sample $+HNO_3 + (NH_4)_2 MoO_4 \rightarrow$ Yellow precipitate on heating
(S)	Phosphate (PO_4^{3-})	4.	Sample $+FeCl_3 \xrightarrow{H_2O}$ deep Red colour $\xrightarrow{\Delta}$ Brown red ppt.

(A) $P \rightarrow 1, Q \rightarrow 2, R \rightarrow 3, S \rightarrow 4$
(B) $P \rightarrow 3, Q \rightarrow 4, R \rightarrow 2, S \rightarrow 1$
(C) $P \rightarrow 2, Q \rightarrow 1, R \rightarrow 4, S \rightarrow 3$
(D) $P \rightarrow 4, Q \rightarrow 3, R \rightarrow 1, S \rightarrow 2$

For More Join: @IITJEE_Advanced

31. Which of the following nickel(II) complex reactions with associated color changes is incorrect?

(A) $\left[Ni(H_2O)_6\right]^{2+}(aq) + en(aq) \rightarrow \left[Ni(H_2O)_4(en)\right]^{2+}(aq) + 2H_2O$

Color change : Green ☐ Pale Blue

(B) $\left[Ni(H_2O)_4(en)\right]^{2+}(aq) + en(aq) \rightarrow \left[Ni(H_2O)_2(en)_2\right]^{2+}(aq) + 2H_2O$

Color change: Pale Blue ☐Blue/Purple

(C) $\left[Ni(H_2O)_2(en)_2\right]^{2+}(aq) + en(aq) \rightarrow \left[Ni(en)_3\right]^{2+}(aq) + 2H_2O$

Color change: Blue/Purple ☐Violet

(D) $\left[Ni(H_2O)_6\right]^{2+}(aq) + 3en(aq) \rightarrow \left[Ni(en)_3\right]^{2+}(aq) + 6H_2O$

Color change : Green ☐ Yellow

32. Which of the following represents the correct order of acidic strength for the given carboxylic acids ?

(A) $ClCH_2COOH > BrCH_2COOH > HCOOH > ClCH_2CH_2COOH$

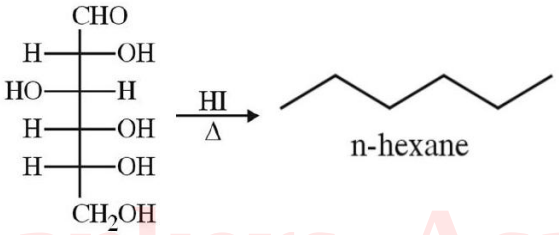
(B) $BrCH_2COOH > ClCH_2COOH > HCOOH > CH_3COOH$

(C) $HCOOH > ClCH_2COOH > BrCH_2COOH > C_6H_5COOH$

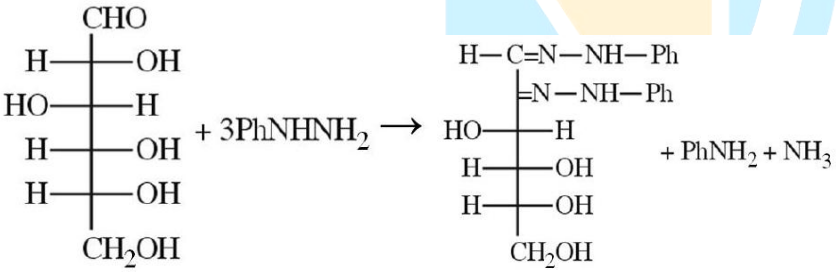
(D) $ClCH_2CH_2COOH > HCOOH > ClCH_2COOH > CH_3CH_2COOH$

33. Which of the following reaction is incorrect?

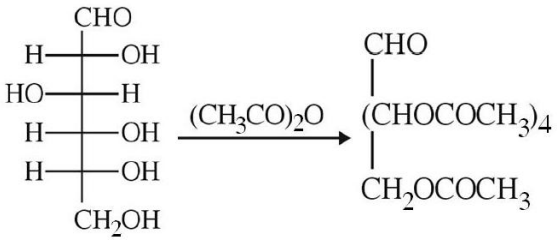
(A)



(B)

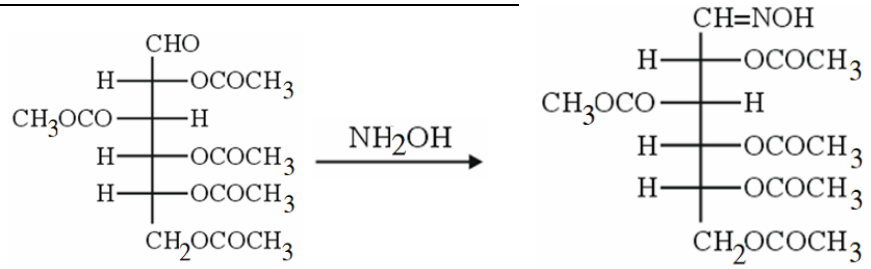


(C)



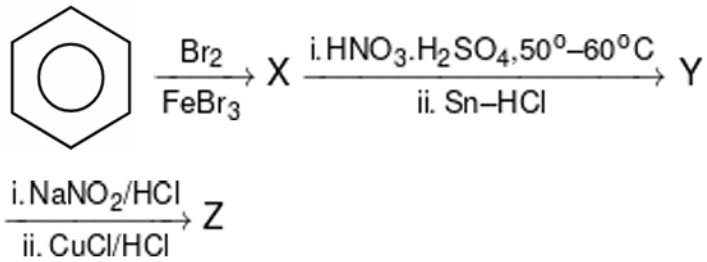
Rankers Academy JEE
THE NARAYANA GROUP

For More Join: @IITJEE_Advanced



(D)

34. Consider the following reaction sequence :



Which of the following correctly identifies compounds X, Y and Z, respectively?

(A) Bromobenzene, 4-bromoaniline, 1-bromo-4-chlorobenzene
(B) Bromobenzene, 3-bromoaniline, 1-bromo-3-chlorobenzene
(C) Chlorobenzene, 4-chloroaniline, 1-chloro-4-bromobenzene
(D) Bromobenzene, 2-bromoaniline, 1-bromo-2-chlorobenzene

35. In the quantum mechanical model of the hydrogen atom, the 1s orbital wave function is given by $\psi(r) = \frac{1}{\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2} e^{-r/a_0}$ where a_0 is the Bohr radius. The probability $|\psi|^2$ indicates the likelihood of finding the electron at a distance r from the nucleus.

Consider the Heisenberg uncertainty principle, which of the following options correctly gives the value of $|\psi|^2$ at $r = a_0$, and explains its significance ?

(A) $\frac{1}{\pi}\left(\frac{1}{a_0}\right)^3 e^{-1}$, showing the electron's exact position.

(B) $\frac{1}{\pi}\left(\frac{1}{a_0}\right)^3 e^{-2}$, representing the probability per unit volume.

(C) $\frac{1}{\pi}\left(\frac{1}{a_0}\right)^{3/2} e^{-2}$, indicating the electron's energy state.

(D) $\frac{1}{\pi}\left(\frac{1}{a_0}\right)^3 e^{-1}$, defining the electron's fixed trajectory

For More Join: @IITJEE_Advanced

36. Which of the following reactions correctly identifies the major product formed under the given conditions?

(A) (S)-3-methylbutan-2-ol reacts with PBr_3 to form (R)-2-bromo-3-methylbutane as the major product.

(B) 2,2-dimethylpropan-1-ol reacts with hot concentrated H_2SO_4 at $140^\circ C$ to form 2-methylbut-1-ene as the major product.

(C) (R)-1-phenylpropan-2-ol reacts with CH_3COCl in the presence of pyridine to form (S)-1-phenylpropan-2-yl acetate as the major product.

(D) 3-methylpentan-3-ol reacts with cold, dilute alkaline $KMnO_4$ to form 3-methylpentane-2,3-diol as the major product.

37. Starting from 1-bromooctane, each sequence below is proposed for the synthesis of octanoic acid. Which sequence(s) will result in octanoic acid as the major product?

A. $1-bromooctane \xrightarrow[(ii)\,B_2H_6,H_2O_2/NaOH]{(i)\,Alc.KOH,\Delta} \xrightarrow{KMnO_4}$

B. $1-bromooctane \xrightarrow[in\ dry\ ether]{Mg} \xrightarrow[H_3O^+]{(i)\,Dry\ ice}$

C. $1-bromooctane \xrightarrow[H_2O]{NaOH} \xrightarrow[H_2SO_4]{CrO_3}$

D. $1-bromooctane \xrightarrow{AgCN} \xrightarrow{H_3O^+}$

Select the **correct** option(s) :-

(A) Only A and B
(B) Only A and C
(C) Only A, B and C
(D) Only B, C and D

38. Which of the following statements about the chemical behavior of amines is **INCORRECT ?**

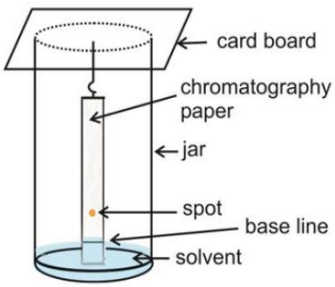
(A) Primary aliphatic amines, when heated with chloroform and alcoholic KOH, produce foul-smelling isocyanides, while secondary and tertiary amines do not undergo this reaction.

(B) Secondary amines react with nitrous acid to form yellow, oily N-nitrosoamines, whereas primary aromatic amines form stable diazonium salts at $0-5^\circ C\,(273-278\ K)$.

(C) Primary amines react with benzenesulphonyl chloride to form sulphonamides that are soluble in alkali due to the presence of an acidic hydrogen on the nitrogen.

(D) Tertiary amines react with benzenesulphonyl chloride to form water-soluble sulphonamide salts under neutral conditions.

39. Which of the following statements about the paper chromatography setup in the figure is **incorrect**?



For More Join: @IITJEE_Advanced

(A) The chromatography paper acts as the stationary phase and contains adsorbed water molecules.

(B) The solvent rises up the paper by capillary action, acting as the mobile phase.

(C) The base line must be submerged in the solvent to ensure the sample spot dissolves for effective separation

(D) The card board cover prevents evaporation of the mobile phase and maintains a saturated atmosphere in the jar.

40. **Statement I :** The metabolism of chloroform in the liver produces phosgene, a highly toxic gas, which contributes to its toxicity, whereas dichloromethane's neurotoxic effects, such as impaired hearing and vision, are primarily due to its interaction with the central nervous system.

Statement II: The storage of chloroform in dark-colored, air-free bottles is critical to prevent its oxidative transformation into phosgene, which is not a concern for dichloromethane due to its lower susceptibility to atmospheric oxidation.

(A) Both Statement I and Statement II are false
(B) Statement I is true but Statement II is false.
(C) Both Statement I and Statement II are true.
(D) Statement I is false but Statement II is true.

41. In the reaction $5A + 3B \rightarrow 4C$, 1.0 mole of A (molar mass $10\ g/mol$) and 0.9 moles of B (molar mass $20\ g/mol$) are dissolved in a solvent to form a 0.5-liter solution. The reaction goes to completion, volume remains constant, then diluted to 2 liters. Another side reaction consumes 20% of C (molar mass $30\ g/mol$) to form an inert byproduct. What is the mass of C in the final solution?

(A) 12 g (B) 19.2 g (C) 24 g (D) 30 g

42. At 950 K, the reaction $Mg(OH)_2(s) \rightleftharpoons MgO(s) + H_2O(g)$ has an equilibrium constant $Kp = 0.36$ atm. A 3.0L sealed vessel initially contains only $Mg(OH)_2(s)$ and $MgO(s)$ with no H_2O gas. After equilibrium is reached, the vessel's volume is reduced to 1.5 L. What is the new equilibrium partial pressure of $H_2O(g)$?

(A) 0.36 atm (B) 0.72 atm
(C) 0.18 atm (D) 0.09 atm

43. A sparingly soluble salt, silver arsenate, Ag_3AsO_4, dissociates in its saturated aqueous solution as follows:

$Ag_3AsO_4(s) \rightleftharpoons 3Ag^+(aq) + AsO_4^{3-}(aq)$

The solubility product (K_{sp}) for silver arsenate at 298 K is 6.4×10^{-22}. Additionally, a solution contains $0.01M Na_3AsO_4$, which completely dissociates to provide AsO_4^{3-} ions. The molar solubility of Ag_3AsO_4 in this solution (in $molL^{-1}$) is :

(A) 1.33×10^{-7} (B) 2.3×10^{-8} (C) 3.6×10^{-9} (D) 4.8×10

44. Which of the following statements is incorrect regarding the reaction kinetics?

(A) The rate of decolorization of potassium permanganate $(KMnO_4)$ in a mixture with oxalic acid $(H_2C_2O_4)$ increases with temperature.

(B) Molecularity is applicable to elementary as well as complex reactions whereas order is applicable

For More Join: @IITJEE_Advanced

only for elementary reactions.

(C) The thermal decomposition of HI on a gold surface is a zero-order reaction.

(D) The reaction $KClO_3 + 6FeSO_4 + 3H_2SO_4 \rightarrow KCl + 3Fe_2(SO_4)_3 + 3H_2O$ is a second-order reaction.

45. Consider the following electrochemical cell at 298 K ;

$Ag(s)\left|Ag^+(0.075M)\right|\left|Hg^{2+}(1.25M)\right|Hg(l)$

The standard reduction potentials are $Hg^{2+}(aq) + 2e^- \rightarrow Hg(l), E^\circ = +0.854\ V$, and

$Ag^+(aq) + e^- \rightarrow Ag(s), E^\circ = +0.800\ V$.

What is the cell potential (E_{cell}) under these conditions, and is the reaction spontaneous as written?

(Use $\frac{RT}{nF} = \frac{0.06}{n}$)

(A) 0.246 V, Spontaneous (B) 0.054 V , Non-spontaneous

(C) -0.119 V , Non-spontaneous (D) 0.124 V , Spontaneous

SECTION-II

(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:

Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. In a reaction with Tollens' reagent, 0.5 moles of propanal (CH_3CH_2CHO) are oxidized completely. Tollens' reagent (ammoniacal silver nitrate) oxidizes aldehydes to carboxylic acids, reducing Ag^+ to metallic silver. Assuming the reaction goes to completion, If the number of moles of silver (Ag) deposited is p , then 100P is

47. 3.0 moles of an ideal monatomic gas undergoes two processes: 1. Isothermal and reversible expansion from 4 L to 8 L at 400 K 2. Then isochoric cooling until the pressure becomes 1/8 of the initial pressure

Calculate the magnitude of total entropy change (ΔS) of the system in J/K.

Given: $R = 8.314\ J/mol\cdot K, C_{v,m} = (3/2)R$ for a monatomic ideal gas, $\ln 2 = 0.7$

48. How many of the following coordination compounds involve inner $(n-1)dx^2 - y^2$ orbitals in their hybridization and exhibit optical isomerism?

i) $[CrCl_2(ox)_2]^{3-}$

ii) $[Co(NH_3)_2Cl_2(en)]^+$.

iii) $[Fe(NH_3)_2(CN)_4]^-$

iv) $[Ni(en)_3]^{2+}$

For More Join: @IITJEE_Advanced

v) $\left[PtCl_2(en)_2\right]^{2+}$

vi) $\left[CoCl_2(en)_2\right]^{+}$

vii) $\left[Co(NH_3)Cl(en)_2\right]^{2+}$

49. The standard molar enthalpy of formation $\left(\Delta_f H^0\right)$ is the enthalpy change when one mole of a compound is formed from its elements in their most stable states at 25^0C and 1 bar pressure. Given the following standard molar enthalpies of formation :

$N_2(g)+2O_2(g)\rightarrow N_2O_4(l);\Delta_f H^o=-20\ kJ\ mol^{-1}$

$C(\text{graphite, s})+2H_2(g)+\frac{1}{2}O_2(g)\rightarrow CH_3OH(l);$

$\Delta_f H^o=-240\ kJ\ mol^{-1}$

$C\ (\text{graphite, s})+2H_2(g)\rightarrow CH_4(g);$

$\Delta_f H^0=-75\ kJ\ mol^{-1}$

Calculate the enthalpy change (ΔH°) for the reaction in $kJmol^{-1}$

$2CH_3OH(l)+N_2O_4(l)\rightarrow 2CH_4(g)+N_2(g)+3O_2(g)$

50. One mole of ionic solid PQ_3 (molar mass $=200\ g/mol$) dissociates into P^{3+} and $3Q^-$ in 1kg of water, giving a boiling point elevation of 2.08 K $\left(K_b=0.52\ K\ kg/mol\right)$. If PQ_3 is 100% ionized in solution then average molar mass of the dissociated solute is ____ $gmol^{-1}$?

MATHEMATICS

MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. $\int\left(3x^2+7x-5\right)\left(x+\frac{1}{x}\right)e^{x+\frac{1}{x}}dx$ is equal to :

(C is constant of integration)

(A) $\left(3x^3+7x^2-5x\right)e^{x+\frac{1}{x}}+C$ (B) $\left(3x^3-7x^3-5x^2\right)e^{x+\frac{1}{x}}+C$

(C) $\left(3x^3-2x^2+5x\right)e^{x+\frac{1}{x}}+C$ (D) $\left(3x^3-2x^2-5x\right)e^{x+\frac{1}{x}}+C$

For More Join: @IITJEE_Advanced

52. Let lines $L_1 : \frac{x-1}{1} = \frac{y-a}{-1} = \frac{z}{2}$ & $L_2 : \frac{x-a}{-1} = \frac{y-1}{2} = \frac{z-2}{1}$. If shortest distance between L_1 & L_2 is $\frac{6}{\sqrt{35}}$ then the sum of possible values of a is :

(A) 2 (B) 4 (C) 6 (D) 8

53. Assertion : - Number of ways in which 75600 can be resolved as a product of two positive divisors is 60

Reason : - If total divisors of N is ' t ' then number of ways in which N can be resolved as a product of two positive divisors is $\frac{t}{2}$.

(A) Both assertion and reason are true and reason is correct explanation of assertion.

(B) Both assertion and reason are true and reason is not correct explanation of assertion.

(C) Assertion is true but reason is false.

(D) Assertion is false.

54. Let the position vectors of the vertices A, B and C of tetrahedron ABCD be $\hat{i}-\hat{j}+\hat{k}, \hat{i}+\hat{j}+2\hat{k}$ and $2\hat{i}+\hat{j}-3\hat{k}$ respectively and D lies on line $L_1 : \vec{r} = \left(10\hat{i}-6\hat{j}+6\hat{k}\right)+\lambda\left(9\hat{i}-5\hat{j}+5\hat{k}\right)$. The altitude from the vertex D to the opposite face ABC meets at the centroid of $\triangle ABC$. The volume of tetrahedron ABCD is equal to :

(A) $\frac{35}{6}$ (B) $\frac{35}{2}$ (C) $\frac{35}{3}$ (D) $\frac{35}{12}$

55. The value of $\frac{\int_0^{\infty} \frac{\ln\left(x+\frac{1}{x}\right)}{1+x^2}dx}{\int_0^{\frac{\pi}{4}} \ln(1+\tan x)dx}$ is equal to :

(A) 2 (B) 4 (C) 6 (D) 8

56. Let R be the region defined by

$R = \left\{(x, y) \in \mathbb{R}^2 : 0 \leqslant x \leqslant 1, (y-x)^2 \leqslant x^3\right\}$.

The area of region R (in sq. units) is :

(A) $\frac{1}{5}$ (B) $\frac{3}{5}$ (C) $\frac{4}{5}$ (D) $\frac{8}{5}$

57. The solution of the differential equation $2x^2 y dy + \left(1-y^2\right)\left(x+y^2-1\right)dx = 0$, ($x > 0$ and c is an arbitrary constant) is :

For More Join: @IITJEE_Advanced

(A) $y^2 = 1 - \frac{x}{c + \ell nx}$ (B) $y^2 = \frac{x}{c + \ell nx}$

(C) $y^2 = x + \frac{c}{x + \ell nx}$ (D) $y^2 = 1 + \frac{c}{x + \ell nx}$

58. $A = \{m \in \mathbb{R} : x^2 - mx + 4 = 0$ has no real roots $\}$ and

$B = \{n \in \mathbb{R} : f(x) = nx^2 + nx + 1 > 0, \forall x \in \mathbb{R}\}$, Then $A \cap B =$

(A) $\{x \in \mathbb{R} : 0 < x < 4\}$ (B) $\{x \in \mathbb{R} : 0 \le x < 4\}$

(C) $\{x \in \mathbb{R} : 0 \le x \le 4\}$ (D) $\{x \in \mathbb{R} : 0 < x \le 4\}$

59. Let $f(x) = \sin x + \int_{\frac{\pi}{2}}^{\frac{\pi}{2}} (\sin x + t\cos x) f(t)\, dt$ then match the entries of List-I with List-II and mark the correct option -

List-I		List-II	
(A)	If range of $f(x)$ is $[\alpha, \beta]$ then $\alpha + \beta$ is	(I)	$\frac{2}{3}$
(B)	$f\left(\frac{3\pi}{2}\right)$ is	(II)	$\frac{1}{\sqrt{2}}$
(C)	$f(\pi)$ is	(III)	$\frac{1}{3}$
(D)	$f\left(\frac{5\pi}{4}\right)$ is	(IV)	o

(A) (A) - III, (B) - I, (C) - IV, (D) – II (B) (A) - III, (B) - II, (C) - I, (D) - IV

(C) (A) - I, (B) - III, (C) - IV, (D) – II (D) (A) - IV, (B) - III, (C) - I, (D) – II

60. The numbers $1, 2, 3, 4, 5, 6$ are arranged around a circle. An arrangement is called valid if the product of every pair of adjacent numbers is at most 15. (Reflection are considered different). The number of valid circular arrangements is

(A) 4 (B) 6 (C) 12 (D) 24

61. Let an ellipse, $E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. The length of the minor axis is equal to one fifth of the distance between its foci. Eccentricity of the ellipse is

(A) $\frac{4}{\sqrt{26}}$ (B) $\frac{\sqrt{26}}{2}$ (C) $\frac{5}{\sqrt{26}}$ (D) $\sqrt{26}$

For More Join: @IITJEE_Advanced

62. If the system of equations

$2x+\lambda y+z=4$

$x+3y-z=2$

$3x+7y+\mu z=6, (\lambda,\mu\in\mathbb{Z})$ has infinitely many solutions, then which of the following value for $(\lambda^2+\mu^2)$ is not possible?

(A) 73 (B) 106 (C) 100 (D) 34

63. Let M be the set of all 3×3 real matrices with entries from $S=\{-1,1\}$. Define $S_1=\{A\in M: A=A^T\}$ and $S_2=\{A\in M: A^T=-A\}$. Then the value of $|S_1|+|S_2|$ is

(Note : $|A|$ denotes number of elements in set A)

(A) 32 (B) 48 (C) 64 (D) 80

64. Let $A=\{(x,y)\in\mathbb{R}^2: y=(\sin^{-1}\sin x)^2\}$ and $B=\{(x,y)\in\mathbb{R}^2: 4y=|x|\}$. Then $|A\cap B|$ is (where $|x|$ denotes number of elements in set x)

(A) 9 (B) 11 (C) 13 (D) 15

65. Statement-1 : $\tan x(2x+\sin 2x)>6x\ \forall x\in\left[0,\frac{\pi}{4}\right)$ and

Statement-2 : If $f(x)$ and $g(x)$ are continuous in $[a,b]$, differentiable in (a,b), $f(a)=g(a)$ and $f'(x)<g'(x)\ \forall x\in(a,b)$, then $f(x)<g(x)\ \forall x\in(a,b)$

(A) Statement-1 is True, Statement-2 is True

(B) Statement-1 is False, Statement-2 is False

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

66. Let z_1,z_2,z_3 be three complex numbers on the circle $|z|=1$ with $\arg(z_1)=-\frac{\pi}{6}, \arg(z_2)=0$, and $\arg(z_3)=\frac{\pi}{6}$.

If $|z_1\bar{z}_2+z_2\bar{z}_3+z_3\bar{z}_1|^2=\alpha+\beta\sqrt{3}, \alpha,\beta\in\mathbb{Z}$ then the value of $\alpha^2+\beta^2$ is equal to

(A) 7 (B) 13 (C) 19 (D) 25

67. In the expansion of $(1+x)^{2n+1}$, let A be the coefficient of the 13th term, B be the coefficient of the 12th term. If $2A=3B$, then number of distinct terms in the expansion of $\left(x^3+1+\frac{1}{x^3}\right)^n$

(A) 29 (B) 15 (C) 31 (D) 43

For More Join: @IITJEE_Advanced

68. A bag has 4 yellow, 3 Blue and 3 Green balls. Balls are drawn one by one without replacement. The probability that the first green ball appears on the 4th draw is

(A) $\frac{1}{6}$ (B) $\frac{1}{8}$ (C) $\frac{3}{20}$ (D) $\frac{2}{15}$

69. The mean of five observations is 4 and their variance is 5.2. If three of these observation are 1,2 and 6 then the other two are

(A) 2 and 9 (B) 3 and 8 (C) 4 and 7 (D) 5 and 6

70. If R and S are two non void relations on a set A . Then, which of the following statements is incorrect?

(A) R and S are transitive implies $R \cap S$ is transitive

(B) R and S are transitive implies $R \cup S$ is transitive

(C) R and S are symmetric implies $R \cup S$ is symmetric

(D) R and S are reflexive implies $R \cap S$ is reflexive

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. Let $A=\{x\in\mathbb{R}:\sin x=\cos 2x\}$ & $B=\{x\in\mathbb{R}:x\in[0,\infty)\cap A\}$. Minimum value of $\sum_{i=1}^{20}x_i=m\pi$, where $x_i\in B$ and $|x_{i+1}-x_i|>1$, then the value of m is :

72. If the function $f(x)=\frac{\sin(\tan x)-\tan(\sin x)}{\sin(\tan(x-\sin x))}$ is continuous at $x=0$, then $f(0)$ is equal to

73. If exhaustive set of values of x satisfying the equation $\cos^{-1}\left(\frac{8x}{1+16x^2}\right)=-\frac{\pi}{2}+2\tan^{-1}(4x)$ is $[a,\infty)$, then $\frac{1}{a}$ is

74. Let m_1 and m_2 be the slopes of the tangents drawn from the point $P(4,1)$ to the hyperbola $H:\frac{y^2}{25}-\frac{x^2}{16}=1$. If Q is the point from which the tangents drawn to H have slopes $|m_1|$ and $|m_2|$ and they make positive intercepts α and β on the x-axis, then $\frac{(PQ)^2}{\alpha\beta}$ is equal to

75. The curve $x+y-\ln(x+y)=2x+5$ has a vertical tangent at the point (α,β). Then $\alpha+\beta$ is equal to :

For More Join: @IITJEE_Advanced